

REPORT

Milk Quality Grade Prediction Using Random Forest Classification

1. Problem Definition

Milk quality assessment is an important task in the dairy industry to ensure food safety and maintain product standards. Manual inspection can be time-consuming and may not always provide consistent results. Therefore, machine learning techniques can be used to automatically classify milk quality based on its physical and chemical characteristics.

Objective

The objective of this project is to develop a machine learning model capable of predicting the quality grade of milk (High, Medium, or Low) using various milk quality parameters.

2. Dataset Description

The dataset used in this project contains milk samples with different quality-related attributes and their corresponding quality grades.

Input Features

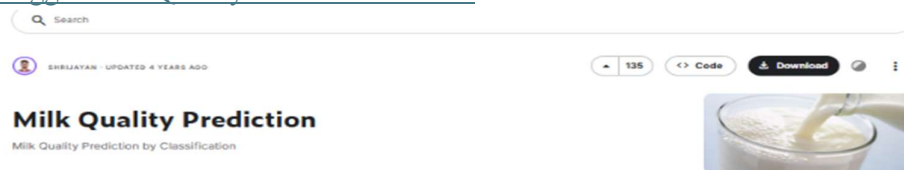
- pH
- Temperature
- Taste
- Odor
- Fat
- Turbidity
- Colour

Target Variable

- Grade (High, Medium, Low)

Dataset Source

[Kaggle Milk Quality Prediction Dataset](#)



3. Data Preprocessing

Several preprocessing steps were performed before training the model.

Handling Missing Values

The dataset was checked for missing values. No significant missing values were found.

Removing Duplicate Records

```
data = data.drop_duplicates()
```

Duplicate entries were removed to improve data quality.

Encoding Categorical Variables

The target variable Grade was converted into numerical values using Label Encoding.

```
encoder = LabelEncoder()
```

```
data["Grade"] = encoder.fit_transform(data["Grade"])
```

Feature Scaling

Feature scaling was performed using StandardScaler to normalize feature values.

```
scaler = StandardScaler()
```

Train-Test Split

The dataset was divided into training and testing sets using an 80:20 ratio.

```
x_train, x_test, y_train, y_test = train_test_split(  
    x, y, test_size=0.2, random_state=42  
)
```

4. Model Building

Selected Model

Random Forest Classifier

Justification

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. It is suitable for classification problems and performs well on datasets with multiple input features.

Model Training

```
model = RandomForestClassifier(  
    n_estimators=100,  
    random_state=42  
)  
model.fit(x_train, y_train)
```

5. Model Evaluation

The performance of the model was evaluated using Accuracy Score.

```
accuracy = accuracy_score(y_test, y_predict)
```

Results

The Random Forest Classifier achieved an accuracy of approximately 82.35294117647058 %.

Interpretation

The obtained accuracy indicates that the model correctly classified milk samples into their respective quality grades. The model demonstrates the capability of machine learning in predicting milk quality based on physical and chemical properties.

6. Conclusion

This project successfully developed a Random Forest Classification model for predicting milk quality grades. Various preprocessing techniques such as duplicate removal, label encoding, feature scaling, and train-test splitting were performed before model training.

The model achieved an accuracy of approximately 82.35294117647058 %, demonstrating satisfactory performance in classifying milk samples into High, Medium, and Low quality categories.

Limitations

- Limited dataset size.
- Prediction accuracy

Future Improvements

- To use a larger dataset.

7. Reflection

This project helped me understand the complete machine learning workflow from data preprocessing to model deployment. I learned how to clean data by removing duplicate records, encode variables using Label Encoding, apply feature scaling, and split the dataset into training and testing sets. I also gained practical experience in implementing the Random Forest Classifier for a real-world classification problem.

One of the challenges faced during the project was understanding the relationship between preprocessing techniques and model performance. I also encountered issues related to feature scaling and model prediction consistency, which helped me understand the importance of maintaining the same preprocessing steps during both training and prediction phases.

Through this project, I learned key machine learning concepts such as classification, feature engineering, model training, evaluation using accuracy metrics, and model persistence using Joblib. The project provided hands-on experience with the complete machine learning pipeline and improved my understanding of how machine learning models can be applied to solve practical problems in food quality assessment.

Code

Program

```
import pandas as pd

from sklearn.ensemble import RandomForestClassifier

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder, StandardScaler

from sklearn.metrics import accuracy_score

import joblib

data = pd.read_csv(r"C:\flutter_projects\INTERNSHIP\Iv1 2\miniproject\milknew.csv")

data = data.drop_duplicates()

#encoders

encoder = LabelEncoder()

data["Grade"] = encoder.fit_transform(data["Grade"])

#data assigning

x = data.drop("Grade", axis=1)

y = data["Grade"]

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)

#standard scaler

scaler = StandardScaler()

x_train = scaler.fit_transform(x_train)
```

```

x_test = scaler.transform(x_test)

#model training

model=RandomForestClassifier(n_estimators=100,random_state=42)

model.fit(x_train,y_train)

#prediction and accuracy

y_predict=model.predict(x_test)

accuracy=accuracy_score(y_test,y_predict)

print("ACCURACY",accuracy*100,"%")

#dumping to joblib

joblib.dump(model, "milk_random_forest.pkl")

joblib.dump(scaler, "scaler.pkl")

joblib.dump(encoder, "grade_encoder.pkl")

print("Model, encoders and scalers saved successfully.")

```

```

milkpred.py > ...
1  import pandas as pd
2  from sklearn.ensemble import RandomForestClassifier
3  from sklearn.model_selection import train_test_split
4  from sklearn.preprocessing import LabelEncoder, StandardScaler
5  from sklearn.metrics import accuracy_score
6  import joblib
7
8  data = pd.read_csv(r"C:\flutter_projects\INTERNSHIP\lvl 2\miniproject\milknew.csv")
9  data = data.drop_duplicates()
10
11  #encoders
12  encoder = LabelEncoder()
13  data["Grade"] = encoder.fit_transform(data["Grade"])
14
15  #data assigning
16  x = data.drop("Grade", axis=1)
17  y = data["Grade"]
18  x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)
19
20  #standard scaler
21  scaler = StandardScaler()
22  x_train = scaler.fit_transform(x_train)
23  x_test = scaler.transform(x_test)
24  #model training
25  model=RandomForestClassifier(n_estimators=100,random_state=42)
26  model.fit(x_train,y_train)
27
28  #prediction and accuracy
29  y_predict=model.predict(x_test)
30  accuracy=accuracy_score(y_test,y_predict)
31  print("ACCURACY",accuracy*100,"%")
32
33  #dumping to joblib
34  joblib.dump(model, "milk_random_forest.pkl")
35  joblib.dump(scaler, "scaler.pkl")
36  joblib.dump(encoder, "grade_encoder.pkl")
37
38  print("Model, encoders and scalers saved successfully.")
39

```

Predict

```
import pandas as pd

import joblib

#loading the files

model= joblib.load("milk_random_forest.pkl")

scaler= joblib.load('scaler.pkl')

encoder= joblib.load("grade_encoder.pkl")

#user inputs

pH = float(input("Enter pH: "))

temperature = float(input("Enter Temperature: "))

taste = int(input("Enter Taste (0/1): "))

odor = int(input("Enter Odor (0/1): "))

fat = int(input("Enter Fat (0/1): "))

turbidity = int(input("Enter Turbidity (0/1): "))

colour = int(input("Enter Colour (240 to 255): "))

#sample input

sample = pd.DataFrame([ {

    "pH": pH,

    "Temperature": temperature,

    "Taste": taste,

    "Odor": odor,

    "Fat ": fat,

    "Turbidity": turbidity,

    "Colour": colour

}])

#transfrm

sample_scaled = scaler.transform(sample)

prediction = model.predict(sample_scaled)

grade = encoder.inverse_transform(prediction)

print("Predicted Grade:", grade[0] , "quality milk.")
```

```

predict.py > ...
1  import pandas as pd
2  import joblib
3
4  #loading the files
5  model= joblib.load("milk_random_forest.pkl")
6  scaler= joblib.load('scaler.pkl')
7  encoder= joblib.load("grade_encoder.pkl")
8
9  #user inputs
10 pH = float(input("Enter pH: "))
11 temperature = float(input("Enter Temperature: "))
12 taste = int(input("Enter Taste (0/1): "))
13 odor = int(input("Enter Odor (0/1): "))
14 fat = int(input("Enter Fat (0/1): "))
15 turbidity = int(input("Enter Turbidity (0/1): "))
16 colour = int(input("Enter Colour (240 to 255): "))
17
18 #sample input
19 sample = pd.DataFrame([[
20     "pH": pH,
21     "Temperature": temperature,
22     "Taste": taste,
23     "Odor": odor,
24     "Fat ": fat,
25     "Turbidity": turbidity,
26     "Colour": colour
27 ]])
28
29 #transform
30 sample_scaled = scaler.transform(sample)
31
32 prediction = model.predict(sample_scaled)
33 grade = encoder.inverse_transform(prediction)
34
35 print("Predicted Grade:", grade[0] , "quality milk.")

```

Output

Program

ACCURACY 82.35294117647058 %

Model, encoders and scalers saved successfully.

```

PS C:\flutter_projects\INTERNSHIP\lv1 2\miniproject> python "C:\flutter_projects\INTERNSHIP\lv1 2\miniproject\milkpred.py"
ACCURACY 82.35294117647058 %
Model, encoders and scalers saved successfully.

```

Predict

Low

Enter pH: 7.4

Enter Temperature: 65

Enter Taste (0/1): 0

Enter Odor (0/1): 0

Enter Fat (0/1): 0

Enter Turbidity (0/1): 0

Enter Colour (240 to 255): 255

Predicted Grade: low quality milk.

```
PS C:\flutter_projects\INTERNSHIP\lv1 2\miniproject> python "C:\flutter_projects\INTERNSHIP\lv1 2\miniproject\predict.py"
Enter pH: 7.4
Enter Temperature: 65
Enter Taste (0/1): 0
Enter Odor (0/1): 0
Enter Fat (0/1): 0
Enter Turbidity (0/1): 0
Enter Colour (240 to 255): 255
Predicted Grade: low quality milk.
```

Medium

Enter pH: 6.8

Enter Temperature: 45

Enter Taste (0/1): 0

Enter Odor (0/1): 1

Enter Fat (0/1): 0

Enter Turbidity (0/1): 0

Enter Colour (240 to 255): 240

Predicted Grade: medium quality milk.

```
PS C:\flutter_projects\INTERNSHIP\lv1 2\miniproject> python "C:\flutter_projects\INTERNSHIP\lv1 2\miniproject\predict.py"
Enter pH: 6.8
Enter Temperature: 45
Enter Taste (0/1): 0
Enter Odor (0/1): 1
Enter Fat (0/1): 0
Enter Turbidity (0/1): 0
Enter Colour (240 to 255): 240
Predicted Grade: medium quality milk.
```

High

Enter pH: 6.6

Enter Temperature: 40

Enter Taste (0/1): 1

Enter Odor (0/1): 0

Enter Fat (0/1): 1

Enter Turbidity (0/1): 1

Enter Colour (240 to 255): 255

Predicted Grade: high quality milk.

```
PS C:\flutter_projects\INTERNSHIP\lv1 2\miniproject> python "C:\flutter_projects\INTERNSHIP\lv1 2\miniproject\predict.py"
Enter pH: 6.6
Enter Temperature: 40
Enter Taste (0/1): 1
Enter Odor (0/1): 0
Enter Fat (0/1): 1
Enter Turbidity (0/1): 1
Enter Colour (240 to 255): 255
Predicted Grade: high quality milk.
PS C:\flutter_projects\INTERNSHIP\lv1 2\miniproject> []
```